

2.11.1 Problems

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

- a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

$$\bar{x} = \frac{3+2+1-1+0}{5} = 1$$

$$\bar{y} = \frac{4+2+3+1+0}{5} = 2$$

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	-2
0	0	-1	1	-2	-2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

- b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8 \quad , \text{ (slope)}$$

$$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 \times 1 = 1.2 \quad , \text{ (intercept)}$$

when X increases by one unit, the average value of Y increases by b_2 units
 when $X=0$, the expected value of Y is b_1

- c. Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

$$\sum_{i=1}^5 x_i^2 = 9 + 4 + 1 + 1 + 0 = 15$$

$$\sum_{i=1}^5 x_i y_i = 12 + 4 + 3 - 1 + 0 = 18$$

$$\therefore \sum (x_i - \bar{x})^2 = 10, \text{ and } \sum x_i^2 - N\bar{x}^2 = 15 - 5 \times 1^2 = 10$$

$$\therefore \sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$$

$$\therefore \sum (x_i - \bar{x})(y_i - \bar{y}) = 8, \text{ and } \sum x_i y_i - N\bar{x}\bar{y} = 18 - 5 \times 1 \times 2 = 8$$

$$\therefore \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$$

- d. Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row.

Calculate the sample variance of y , $s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $s_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile, of x ?

$$\hat{y}_i = 1.2 + 0.8x_i, \quad \hat{e}_i = y_i - \hat{y}_i$$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$s_y^2 = \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N-1} = \frac{4+0+1+1+4}{4} = 2.5$$

$$s_x^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2.5$$

$$s_{xy} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{N-1} = \frac{8}{4} = 2$$

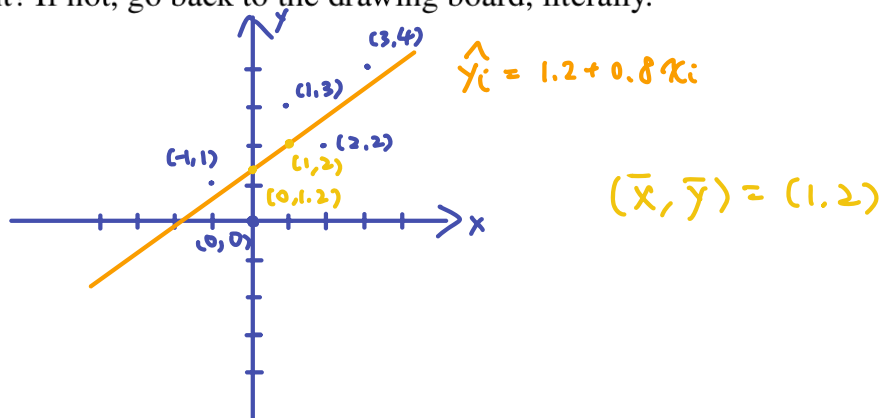
$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2.5} \times \sqrt{2.5}} = 0.8$$

$$CV_x = 100 \left(\frac{s_x}{\bar{x}} \right) = 100 \times \left(\frac{\sqrt{2.5}}{1} \right) \div 158.11$$

median of $x = 1$

- e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_0 x_i$.

- f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.



g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

$$\therefore \bar{y} = 2, \text{ and } b_1 + b_2\bar{x} = 1.2 + 0.8 \times 1 = 2$$

$$\therefore \bar{y} = b_1 + b_2\bar{x}$$

h. Show that for these numerical values $\hat{y} = \bar{y}$, where $\hat{y} = \sum \hat{y}_i / N$.

$$\therefore \hat{y} = \sum \hat{y}_i / N = 10 / 5 = 2, \text{ and } \bar{y} = 2$$

$$\therefore \hat{y} = \bar{y}$$

i. Compute $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^5 \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

j. Compute $\widehat{\text{var}}(b_2|\mathbf{x})$ and $\text{se}(b_2)$

$$\widehat{\text{var}}(b_2|\mathbf{x}) = \frac{\hat{\sigma}^2}{\sum_{i=1}^5 (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$$

$$\text{se}(b_2) = \sqrt{0.12} \doteq 0.3464$$